AIMD-HF: A Minimal, Transparent, and Interactive Educational Implementation of Hartree–Fock Ab Initio Molecular Dynamics with Real-Time OpenGL Visualization

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Abstract

This repository contains a fully self-contained, educational implementation of real-time *Ab Initio Molecular Dynamics* (AIMD) using the Restricted Hartree–Fock (RHF) method with the minimal STO-3G basis set. The simulator supports multiple atomic systems, including hydrogen clusters (H_n), helium atoms (He_n), and protonated helium hydride systems (HeH^+), enabling the exploration of both neutral and charged quantum systems. The code strongly prioritizes **conceptual clarity and pedagogical value** over computational performance. Every fundamental step of quantum chemistry and molecular simulation is implemented explicitly and clearly commented in a single C++ file.

Introduction

AIMD-HF is an interactive educational simulator that couples a full quantum-mechanical electronic structure calculation (Restricted Hartree–Fock) with classical nuclear motion in real time. Unlike conventional molecular dynamics packages that rely on pre-fitted force fields, this program computes the energy and forces *on-the-fly* by solving the electronic Schrödinger equation approximately via the Hartree–Fock method at every molecular dynamics step. This allows users to directly visualize the intimate relationship between electronic structure and atomic dynamics. The inclusion of helium and HeH^+ systems enables exploration of weak interactions, charge asymmetry, and thermally driven dissociation processes.

Key Features

- **Multi-system support:**
  - Hydrogen clusters (H_n)
  - Helium atoms (He_n)
  - Protonated helium hydride systems (HeH^+)
- Complete Restricted Hartree–Fock (RHF) implementation with STO-3G basis set
- Explicit evaluation of all one- and two-electron integrals:
  - Overlap (S)
  - Kinetic energy (T)
  - Nuclear attraction (V_{ne})
  - Electron repulsion integrals (ERI)
- Self-Consistent Field (SCF) procedure with adaptive damping and Jacobi diagonalization

- Nuclear forces computed via central finite differences
- Velocity Verlet integrator
- Langevin thermostat (friction + stochastic noise)
- Berendsen barostat for pressure control
- Reflective (elastic) boundary conditions in a cubic simulation box
- Real-time 3D visualization using OpenGL and GLUT
- Dynamic chemical bond visualization (yellow cylinders)
- Real-time rolling graph of Hartree–Fock energy
- Comprehensive Heads-Up Display (HUD) showing physical observables
- **Interactive Educational Panels (zero extra SCF cost):**
  - **[O]** Molecular Orbital Diagram (energies + occupancy)
  - **[M]** Mulliken Population Analysis (gross population + net charge)
  - **[E]** Energy Decomposition ($T + V_{\mathrm{ne}} + V_{\mathrm{ee}} + V_{\mathrm{nn}}$)
  - **[F]** Real-time quantum force vectors (3D red arrows)
  - **[V]** 3D semi-transparent orbital volume cloud (molecular orbital lobes)
- Multi-threaded design: quantum calculations run in a background thread
- Mutex-protected render buffers for smooth visualization

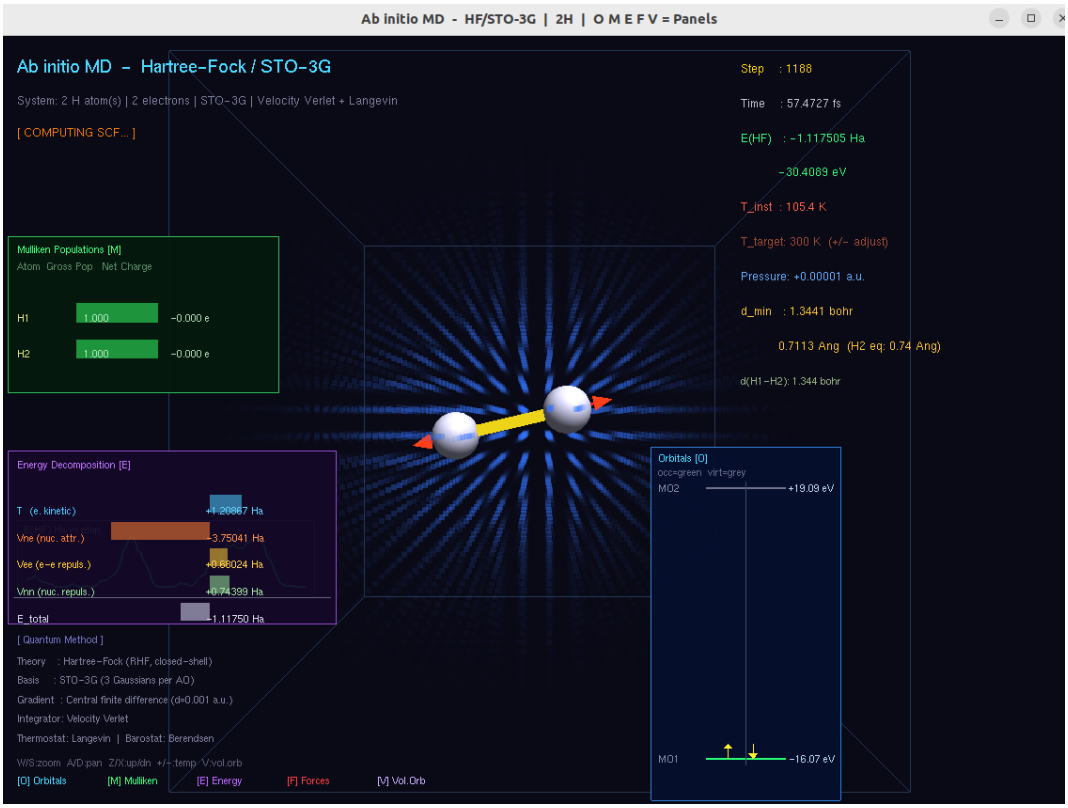

Figure 1: Example of the real-time 3D visualization in AIMD-HF showing atoms (spheres), chemical bonds (yellow cylinders), force arrows (red, when enabled with key F), and the new semi-transparent orbital volume cloud (colored point cloud representing the phase and density of occupied molecular orbitals). The image illustrates a H_2 system with the orbital volume renderer (key V) active. Controls: W/S (zoom), A/D (pan), Z/X (up/down), +/- (temperature), O (orbitals), M (Mulliken), E (energy), F (forces), V (volume cloud).

Installation and Compilation

Linux

On most Linux distributions, the required OpenGL and GLUT libraries can be installed via the package manager. For Debian/Ubuntu-based systems:

```
sudo apt update
sudo apt install build-essential freeglut3-dev libglu1-mesa-dev
```

For Fedora:

```
sudo dnf install gcc-c++ freeglut-devel mesa-libGLU-devel
```

Compilation:

```
g++ aimd_hf.cpp -o aimd_hf -lGL -lGLU -lglut -lm -lpthread -O2
```

Execution:

```
./aimd_hf
```

Windows

On Windows, it is recommended to use MinGW or MSYS2 for compilation. **Step 1: Install MSYS2** Download from: `https://www.msys2.org` **Step 2: Install required packages**

```
pacman -S mingw-w64-x86_64-gcc
pacman -S mingw-w64-x86_64-freeglut
```

Step 3: Compile (using MinGW shell)

```
g++ aimd_hf.cpp -o aimd_hf -lfreeglut -lopengl32 -lglu32 -O2
```

Step 4: Run

```
./aimd_hf
```

Execution Parameters

The program supports multiple systems via command-line:

```
./aimd_hf h              2 hydrogen atoms (default  H  )
./aimd_hf h N
./aimd_hf he
./aimd_hf he N
./aimd_hf heh+
./aimd_hf heh+ N
```

Notes:

- `N` = number of atoms or pairs
- Default system is H_2
- **For hydrogen systems (H_n), N must be even (closed-shell RHF)**
- Charged systems (HeH^+) are fully supported
- Electron count is automatically adjusted

Keyboard Controls

Key	Action
W / S	Zoom in/out
A / D	Camera left/right
Z / X	Camera up/down
+ / -	Temperature ±50 K
O	Molecular orbital diagram
M	Mulliken populations
E	Energy decomposition
F	Force arrows (3D)
V	3D semi-transparent orbital volume cloud
ESC	Exit

Educational Value

This project was designed as a teaching tool for quantum chemistry and molecular simulation.

Users can explore:

- SCF convergence during real-time dynamics
- Coupling between electronic structure and nuclear motion
- Temperature effects on molecular behavior
- Charge redistribution (Mulliken populations)
- HOMO–LUMO evolution
- Energy component contributions
- Quantum mechanical forces acting on atoms

- Shape, phase (sign) and spatial distribution of molecular orbitals in 3D via the orbital volume renderer

Supported Systems and Physics

- H_n: covalent bonding and many-body effects
- He_n: weak interactions (no true bonding in RHF/STO-3G)
- HeH^+: ionic bonding and charge asymmetry

Design Philosophy

- Clarity over performance
- Explicit implementation of all algorithms
- Minimal dependencies
- Strong pedagogical focus

Limitations

- $\mathcal{O}(N^4)$ scaling
- Numerical forces (finite differences)
- Limited to small systems ($\sim$12 atoms)
- Minimal basis (STO-3G)
- Only H and He supported

Supporting Information

The source code is available at: `https://github.com/gallactuz/AIMD-HF`